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Gerda Stetter Stiftung
Technik *macht* Spaß!

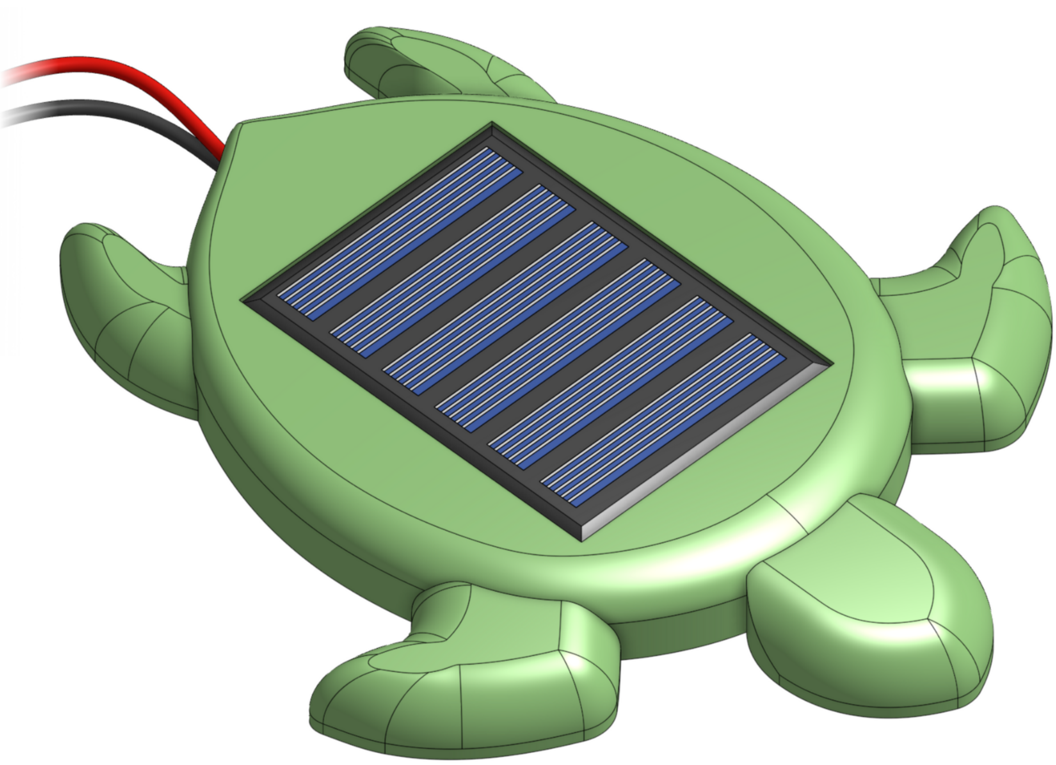
EDUDEMOS

EDUcating through Sustainable **DEMO**nstrators

Assembly

Solar Turtle

Guide for assembly during a workshop.



Find more at
edudemos.eu/en/

Our activities at
technikmachtspass.org

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V 12.25

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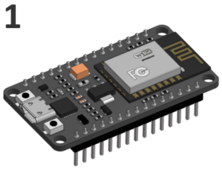
FUNDACIÓN SERGIO ALONSO

Finnova

I.E.S. El Rincón

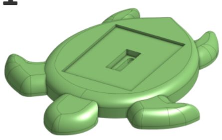
gbs sg.ch

Part-List

1 

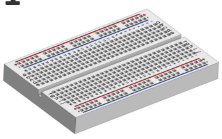
The Microcontroller

... is a small computer. It processes the voltage values of the Solar Panel and sends commands to the Screen.

1 

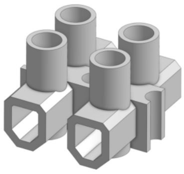
The Turtle

... is used to mount the Solar Panel.

1 

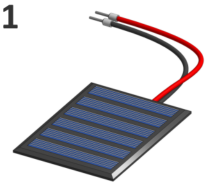
The Breadboard

... makes it easy to connect different components and wires.

1 

The Luster Terminal

... lets you connect wires together.

1 

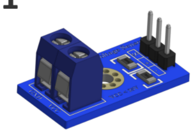
The Solar Panel

... generates electricity when light shines on it. The more light, the more voltage.

1 

The U-Bridge

... is a very short wire for connecting two holes on the Breadboard. Its color has no effect on it.

1 

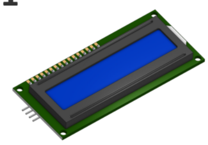
The Voltage Sensor

... divides the solar-voltage by 5. Otherwise it would be too high for measuring in the Microcontroller.

7 

The M-F Wire

... has a pin and a socket. It is used to connect two components. Its color has no effect on it.

1 

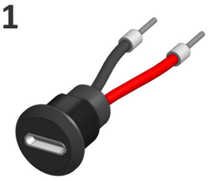
The Screen

... shows the generated voltage of the Solar Panel.

2 

The M-M Wire

... has two pins. It is used to connect two components. Its color has no effect on it.

1 

The USB-C Port

... provides electricity for the entire circuit.

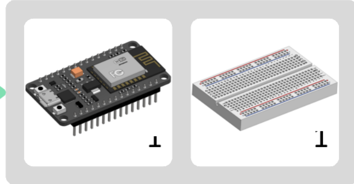
1 

The Screwdriver

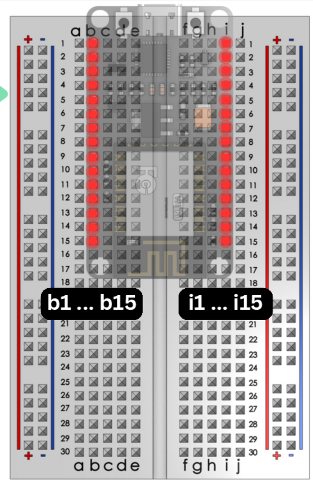
... is used for screwing or unscrewing screws.

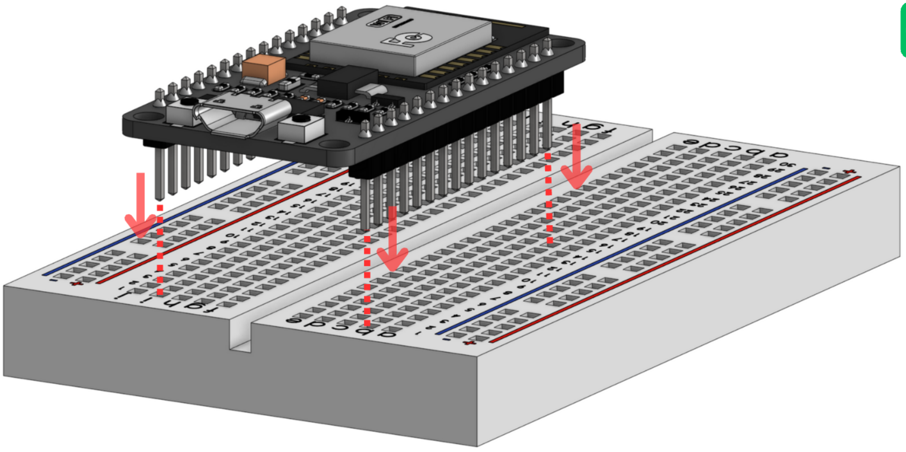
1.1 The Microcontroller

Parts and tools for this step.



Pin-View

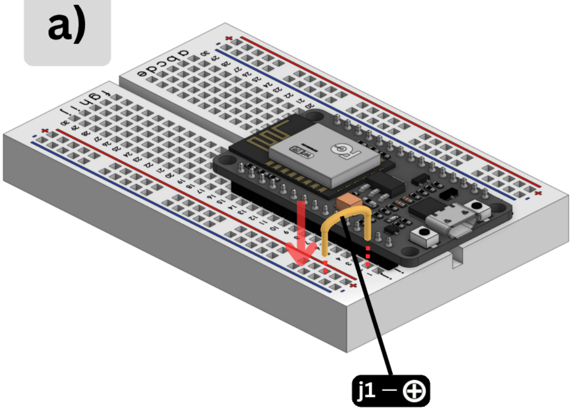




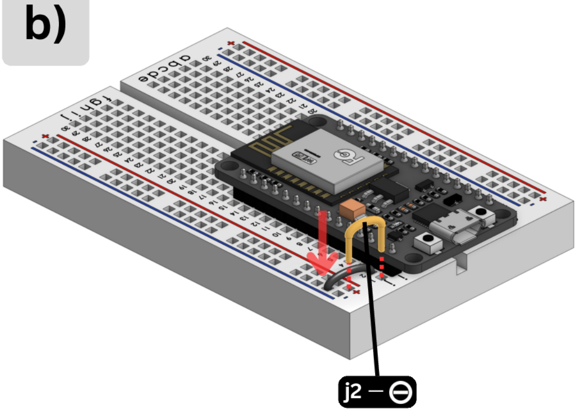
1.2 The U-Bridges

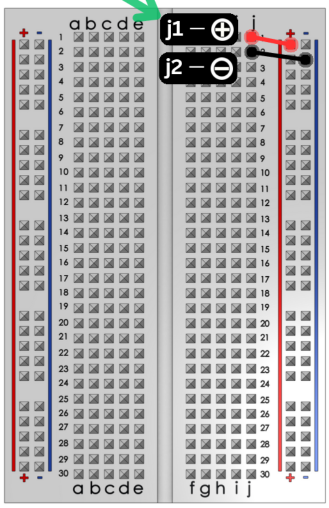
Connect the hole j1 with ⊕ on the Breadboard.

a)

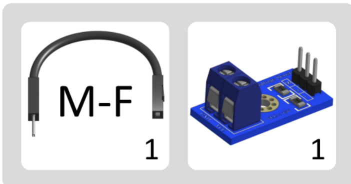


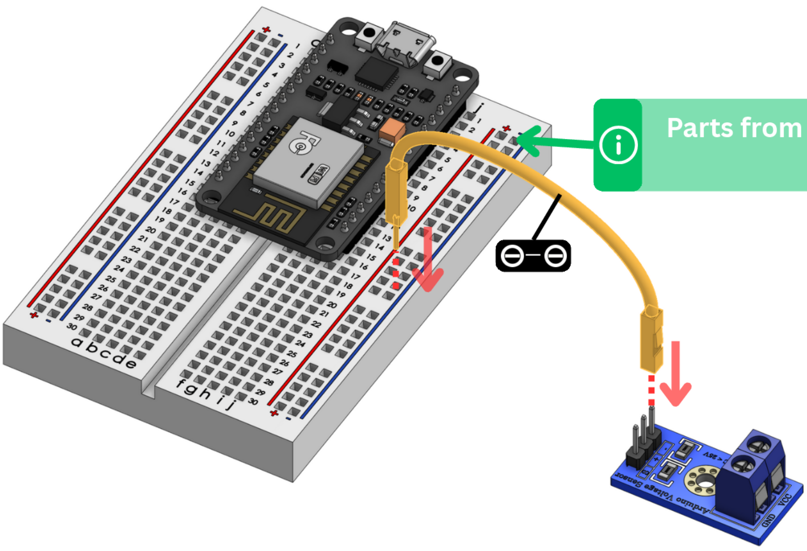
b)



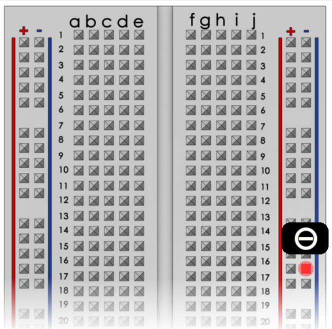


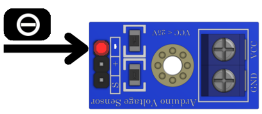
2.1 The Voltag Sensor: The Minus-Wire



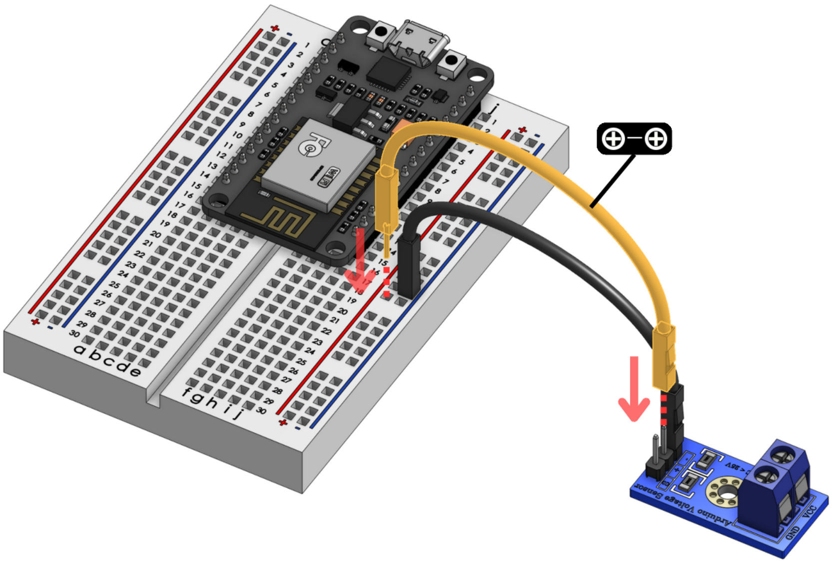


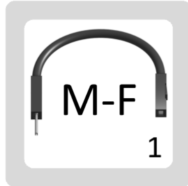
Parts from previous steps will not be shown.

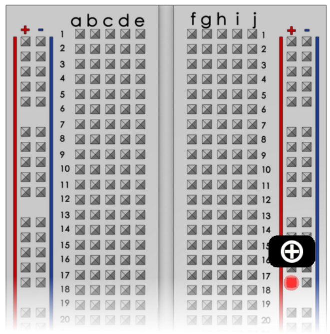


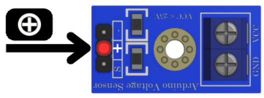


2.2 The Voltage Sensor: The Plus-Wire

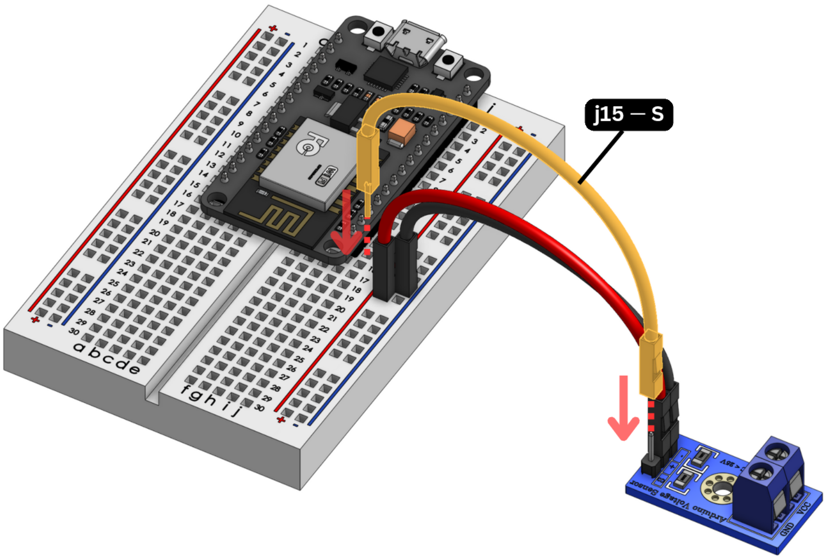


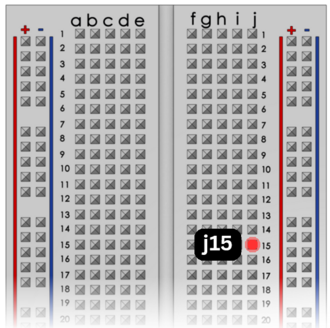


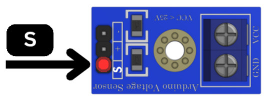




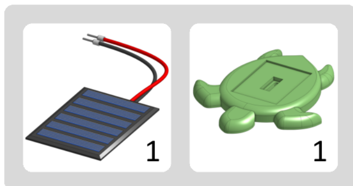
2.3 The Voltage Sensor: The Signal-Wire

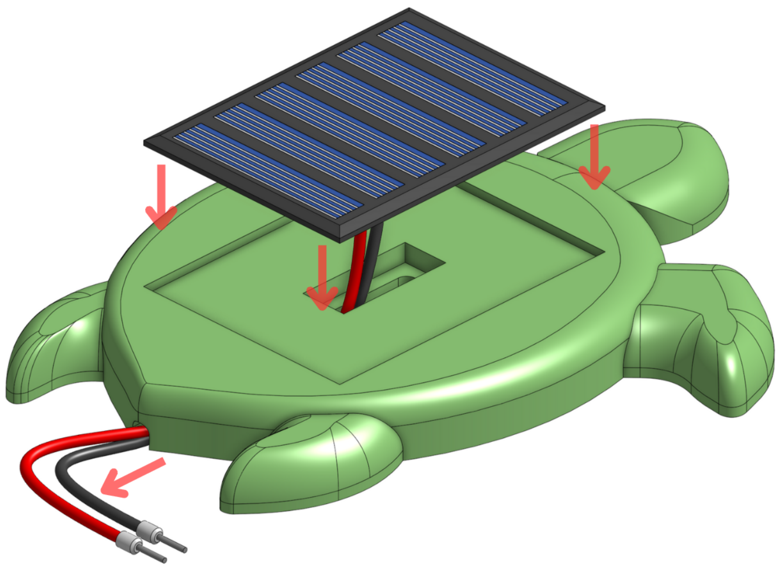






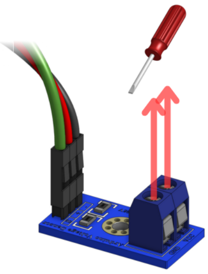
2.4 The Solar Panel and the Turtle



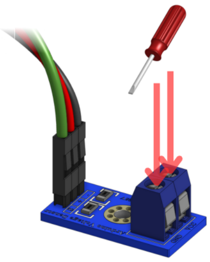


2.5 The Solar Panel and the Voltage Sensor

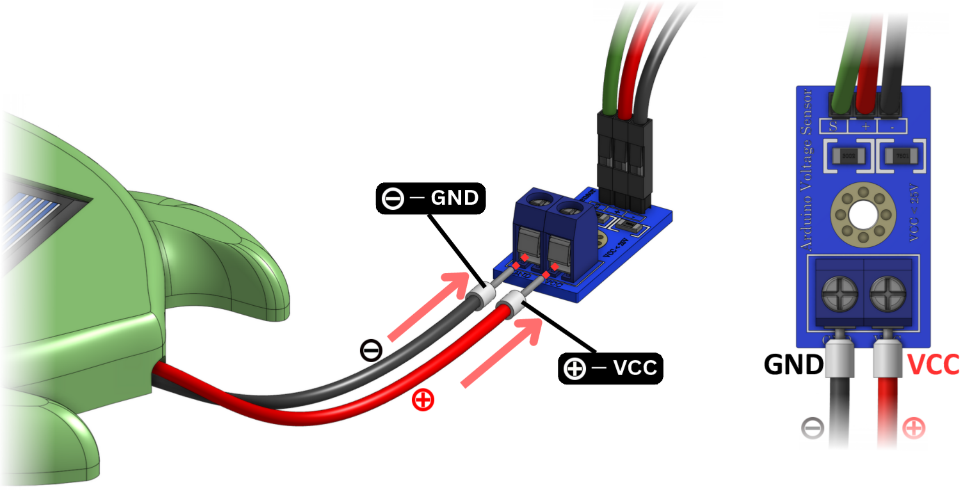
a)



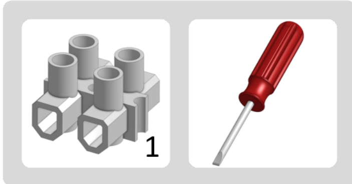
c)



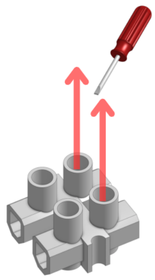
b)



3.1 The USB-Port and the Luster Terminal



a)



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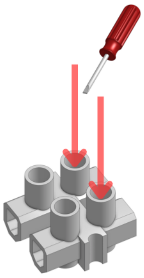

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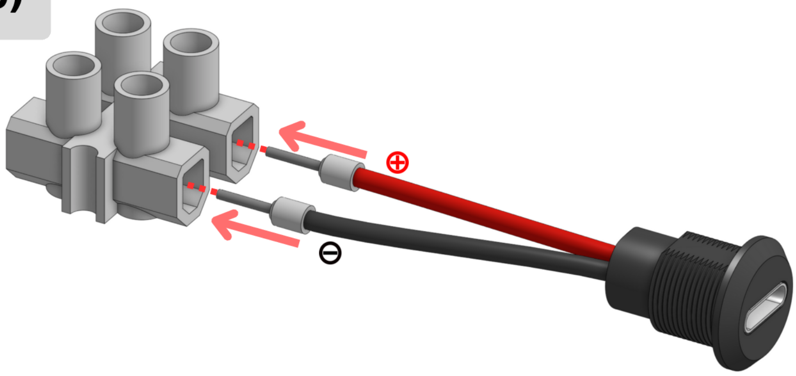
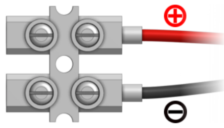
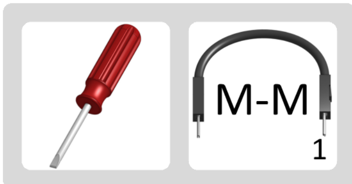
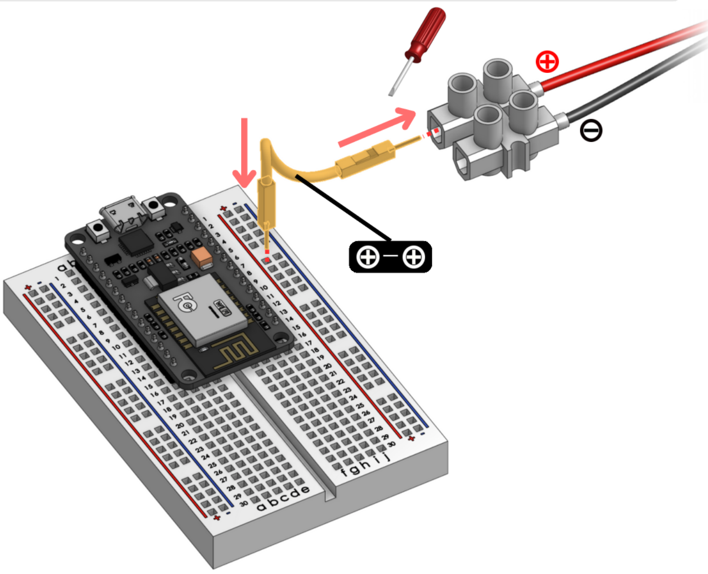
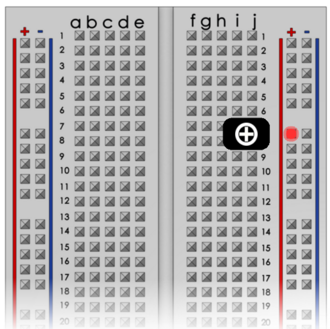


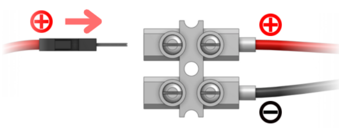


3.2 The Luster Terminal: The <u>Plus</u>-Wire

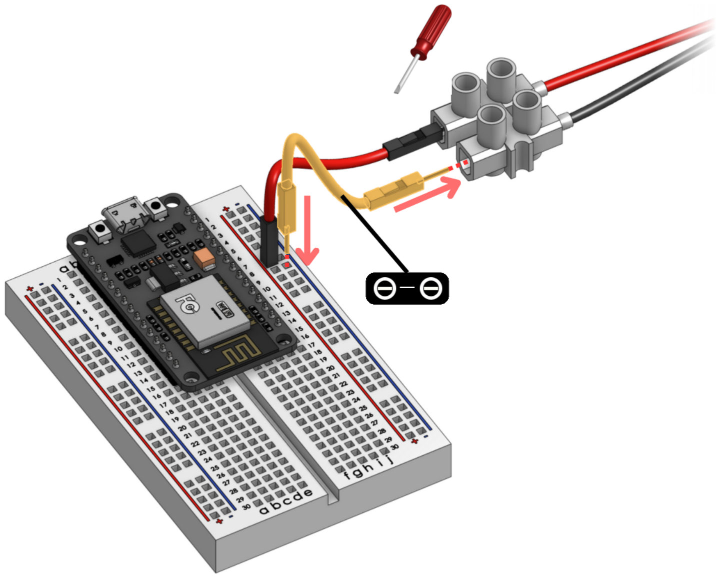


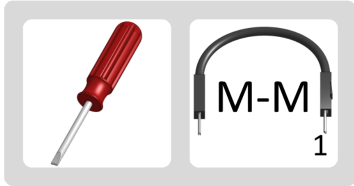


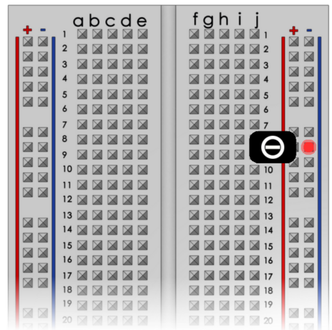


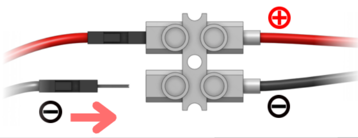


3.3 The Luster Terminal: The Minus-Wire

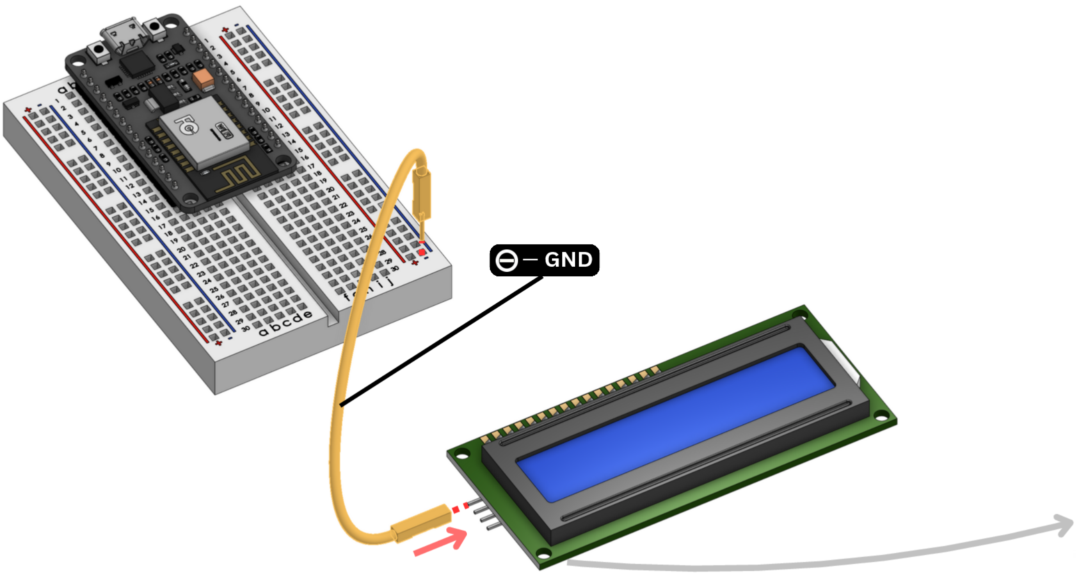


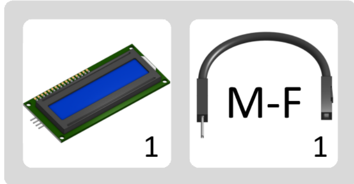


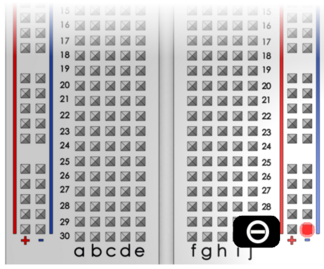


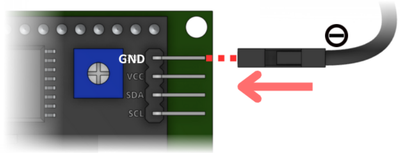


4.1 The Screen: The Minus-Wire

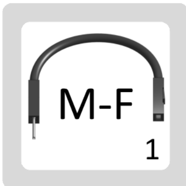


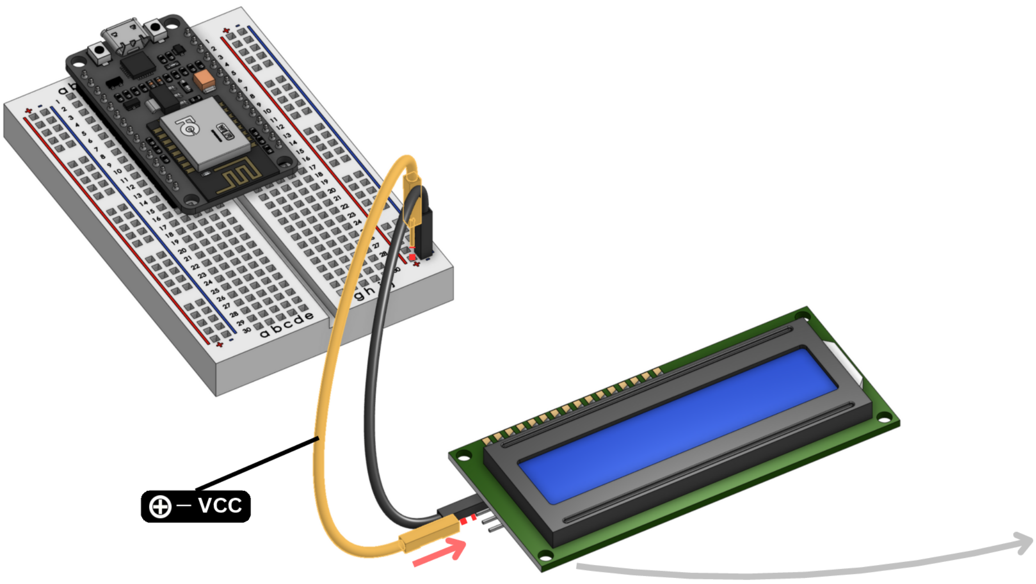


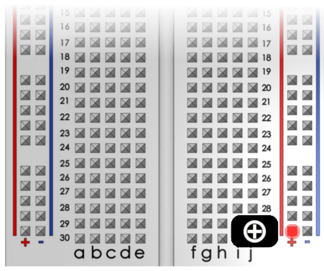


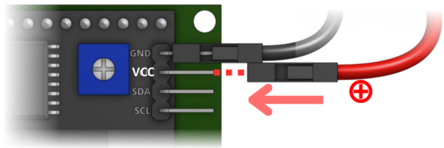


4.2 The Screen: The Plus-Wire

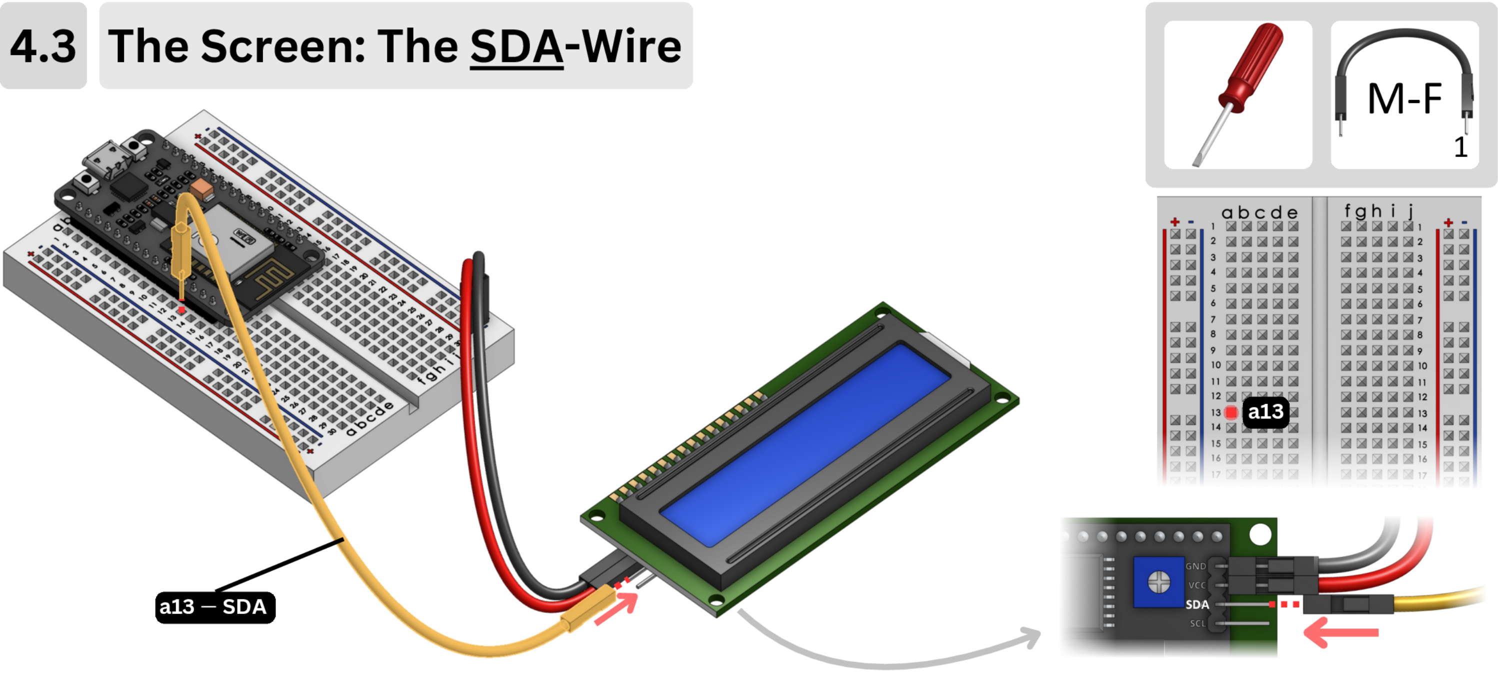




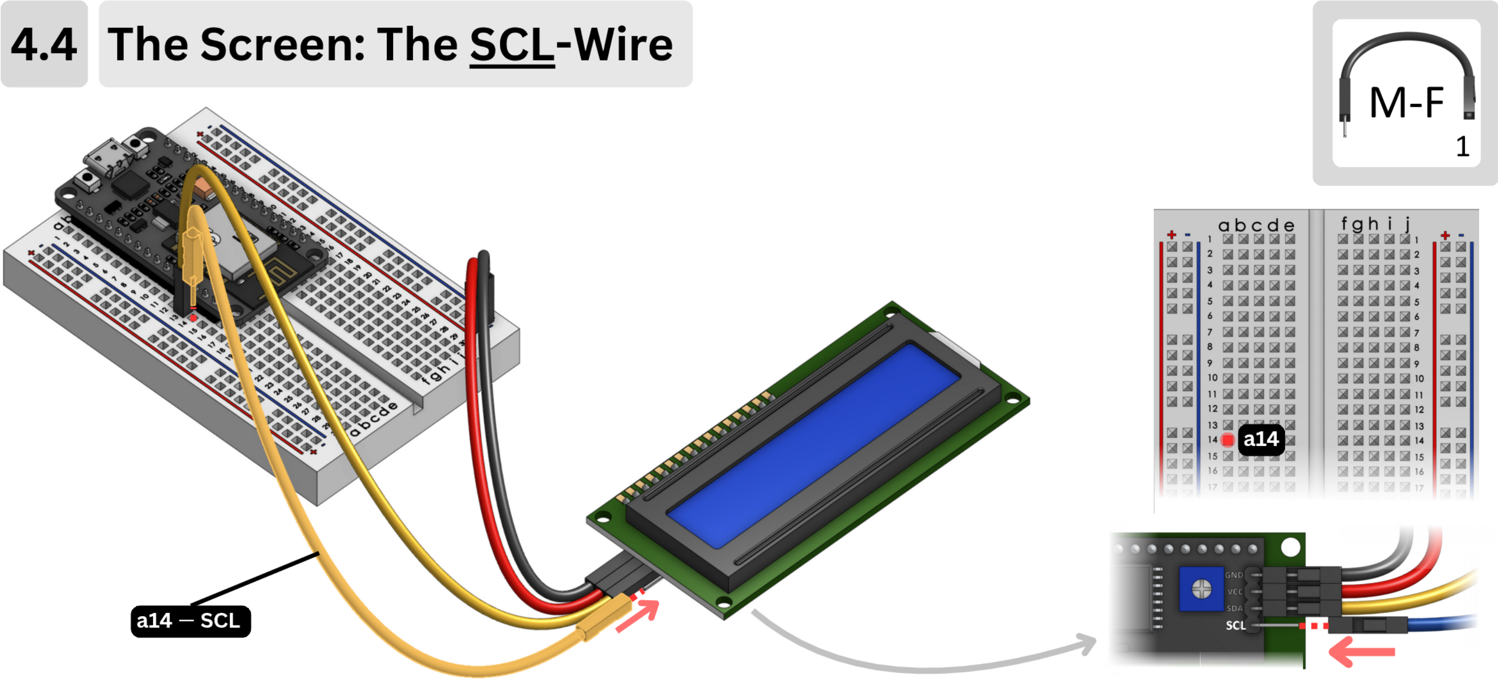




4.3 The Screen: The SDA-Wire

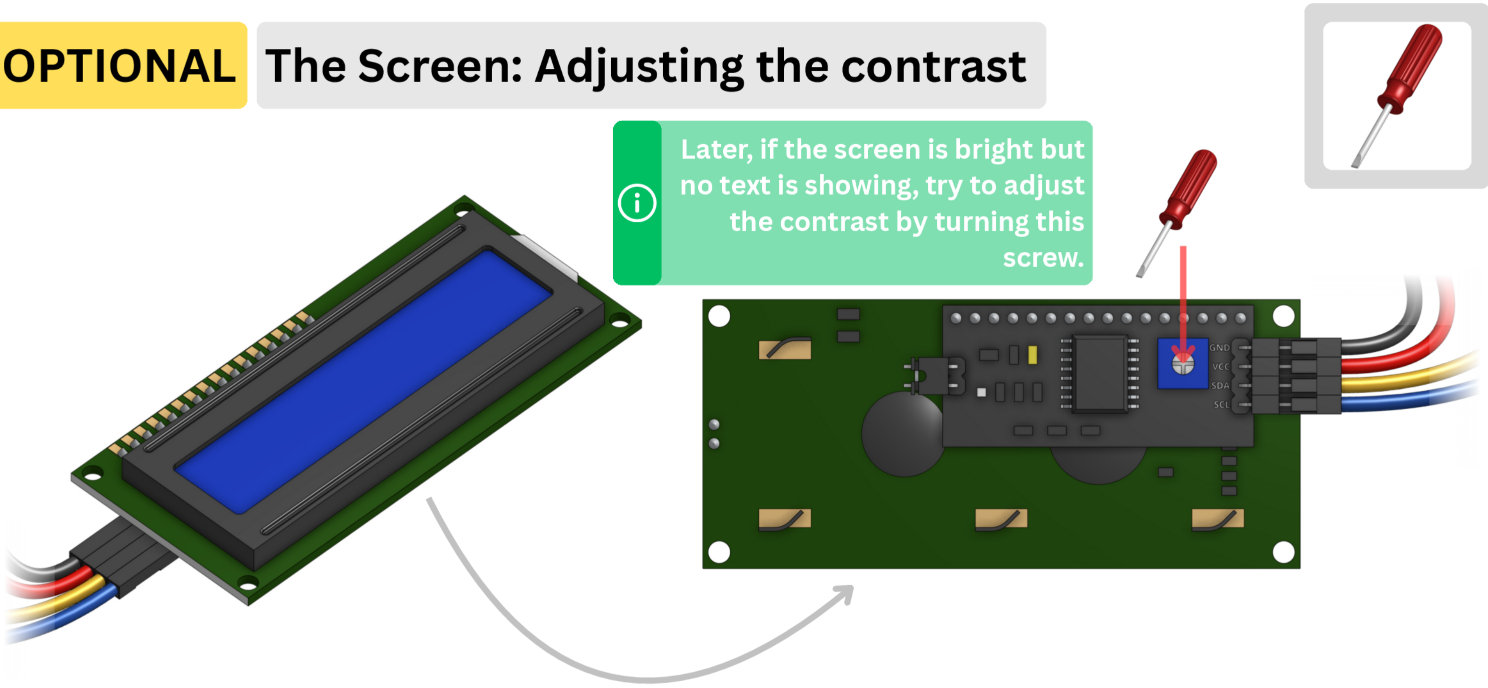


4.4 The Screen: The SCL-Wire



OPTIONAL The Screen: Adjusting the contrast

Later, if the screen is bright but no text is showing, try to adjust the contrast by turning this screw.



Done!